MONOGATE ELECTRONICS / TRAINER BOARD V0

Reflex Lab 01 Quick Start

One EML kernel -> verifiable decisions -> physical output. Start with a potentiometer, LED, and ESP32.

```
potentiometer -> threshold_reflex_v0 -> guard clamp -> LED -> trace -> replay
```

Parts

- ESP32 DevKit or ESP32 ESP-32S breakout
- Breadboard and jumper wires
- 10k potentiometer
- LED plus 220/330 ohm resistor
- USB data cable

Wire It

1. ESP32 GND -> blue/- rail
2. ESP32 3V3 -> red/+ rail
3. Pot outer -> red/+ rail
4. Pot wiper -> GPIO34
5. Pot outer -> blue/- rail
6. GPIO25 -> resistor -> LED anode
7. LED cathode -> blue/- rail

Validate Before Hardware

```
python tools\validate_kernel_spec.py kernels\threshold_reflex_v0
python tools\validate_trace.py kernels\threshold_reflex_v0\traces\golden_trace.jsonl
python tools\replay_trace.py kernels\threshold_reflex_v0\traces\golden_trace.jsonl
```

Upload And Monitor

```
arduino-cli upload -p COM3 --fqbn esp32:esp32:esp32 kernels\threshold_reflex_v0\esp32\threshold_reflex_v0
arduino-cli monitor -p COM3 -c baudrate=115200
```

Safety

- Use 3.3V only.
- Never connect the pot to VIN/5V.
- Always keep the resistor in series with the LED.
- If anything gets warm, unplug USB immediately.

Evidence

- Breadboard photo
- `live_trace.jsonl`
- Validation output
- Replay output
- Short `FINDINGS.md`